

State of the Kryptos Solution

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This is the current state of the solution of Kryptos as of November 18, 2003. The opinions expressed are mine. Others in this group might disagree, so use this information at your own risk.

This document assumes some understanding of earlier work on Kryptos solutions. If you are unfamiliar with Kryptos, additional information may be found in the Yahoo Kryptos Group's *Files* and/or *Links* sections. The **KryptosBackground.pdf** file in the **Gary's PDFs** folder will provide you with transcripts of the complete text and the initial solutions for the first 3 parts.



Note that the information in this document supercedes that in my earlier PDFs. For example, my pet theory about the “errors” being parts of an underlying document showing through, while interesting, now no longer appear as likely. Also the transcripts of the table portion of Kryptos in the earlier PDFs do not include the extra L in the GHI... alphabet that was recently brought to the attention of the group by Richard Anderson.

Another excellent resource is www.elonka.com/kryptos/. Here you will find a very extensive compilation of material on Kryptos and Sanborn's work in general.

A Change in Perspective

Until now all previous solutions to the first 3 parts of Kryptos have been the result of “cracking” the code. They may have been either pencil-and-paper solutions, or computer-aided, but they all relied on extracting the plain text from the cipher text through the use of clever techniques and plain old hard work.

In late October 2003 something happened to change my perspective. Elonka Dunin (*co-moderator of the Yahoo Kryptos Group*) and I had an opportunity to spend some time with both Jim Sanborn (*creator of Kryptos*) and Ed Scheidt (*former head of cypto for the CIA, as well as Sanborn's advisor on Kryptos*).

Both were, as expected, very tight-lipped on the subject of Kryptos. But, during conversations that spanned several days, a few things were mentioned, almost in passing, that changed the way I thought about Kryptos.

Ed Scheidt talked about the differences between the older codes used in Kryptos and modern codes. He focused on the problem of delivering the “key” to an operative in the field. He pointed out that in modern “digital” codes the algorithm is known to everyone. The “key” is simply a long number. In older “analog” codes the “key is the algorithm.” The question he posed: “So how do

you deliver the key in a way that your operative can recover it, and still keep the secret from falling into the wrong hands?" The implication was that the latter condition could be met through the use of intentional dead ends, but he provided no hints regarding the actual delivery of the algorithm.

This prompted me to look at Kryptos from a slightly different perspective – not as simply an encrypted message, but also as a vehicle with which to deliver the key. This changes the task from one of “cracking” a code to one much more like “solving” a puzzle.

Jim Sanborn, talking to Elonka over dinner, pointed out that no one had solved Part 3 using the same matrix he had used to encrypt it.

This was particularly interesting because the Kryptos Group had discovered multiple techniques for solving the transposition. If a clue could be found in Kryptos that would point to one method as being the single correct solution, it could provide valuable information regarding the very nature of Kryptos; i.e., was it an encrypted message to be “cracked,” or a puzzle to be “solved?” As Part 3 also appeared to be the simplest section, it seemed to be the logical place to begin looking for hidden clues.

Recovering the Key and Solving Part 3



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The raised Y, A, and R shown in the above photo were first brought to my attention by Elonka Dunin in April or May of this year. She had noticed them during a visit to the CIA to present a talk on steganography. The CIA allowed her to do rubbings, which she later posted to her Kryptos web site. Since this anomaly occurs at the beginning of Part 3, it seemed likely that it was related to that section. The task was to work backwards from the known solution and find information hidden in this clue that could provide an operative in the field with the “key” – keeping in mind that intentional dead ends are also required to confuse others.

After a bit of trial and error, I was able to construct the following binary-number based solution:

Part 3 begins with ENDYAHROHN. The 5 letter DYAHR grouping defined by the extra spacing before D and after R indicates that each binary number is 5 digits long. The raised letters Y, A, & R represent 1. The other letters are 0. *(Note that the first number reads left to right while the second reads right to left. This “mirroring” could result in the recovery of the wrong number and thus could be one of the intentional dead ends.)*

E	N	D	Y	A		H	R	O	H	H
0	0	0	1	1		0	1	0	0	0
1	2	4	8	16		16	8	4	2	1
24						8				

Write Part 3 into in to a grid of **24 rows** by 14 columns, starting in the upper right corner and writing down the column:

1	D	A	E	G	I	F	W	Q	E	W	R	N	D	E
2	M	G	G	R	B	I	E	H	O	N	A	O	Y	N
3	R	T	O	I	S	O	N	E	E	D	R	H	S	D
4	I	D	E	E	T	A	E	T	I	P	S	H	Y	
5	P	H	A	W	D	U	T	N	F	T	E	N	N	A
6	F	A	O	F	D	E	A	C	O	E	S	O	A	H
7	E	R	S	E	N	T	M	T	L	E	L	S	I	R
8	I	D	D	B	I	U	A	A	S	N	N	M	A	O
9	M	P	D	A	A	A	T	Y	E	R	N	R	C	H
10	E	N	R	E	A	E	E	C	D	A	E	W	H	N
11	H	E	Y	C	H	O	G	D	T	H	L	X	T	L
12	N	O	D	T	T	T	Y	E	I	C	E	M	N	S
13	L	H	L	D	T	O	E	I	W	T	B	N	R	R
14	S	M	O	D	M	A	E	F	E	E	L	E	E	H
15	S	G	R	H	S	R	R	T	N	N	P	T	Y	E
16	T	F	I	I	T	M	L	B	H	E	I	P	U	O
17	T	M	T	L	E	A	B	R	A	U	I	R	L	C
18	R	F	R	C	W	E	T	W	E	D	A	N	D	P
19	T	E	K	E	P	E	E	P	I	R	C	G	S	T
20	V	U	L	I	I	R	E	A	O	E	A	A	L	E
21	D	H	M	H	E	T	F	M	Y	T	E	T	L	O
22	O	E	L	S	R	N	O	H	T	N	W	I	S	I
23	H	E	E	I	O	R	A	H	E	H	M	H	L	B
24	W	C	H	T	A	T	S	E	Y	A	T	N	L	I

Take the text out of the above grid by rows, from left to right. Transfer it to a grid of **8 rows** by 42 columns, again starting at the upper right corner and writing down the column.

1	S	L	O	W	L	Y	D	E	S	P	A	R	A	T	L	Y	S	L	O	W	L	Y	T	H	E	R	E	M	A	I	N	S	O	F	P	A	S	S	A	G	E	D
2	E	B	R	I	S	T	H	A	T	E	N	C	U	M	B	E	R	E	D	T	H	E	L	O	W	E	R	P	A	R	T	O	F	T	H	E	D	O	O	R	W	A
3	Y	W	A	S	R	E	M	O	V	E	D	W	I	T	H	T	R	E	M	B	L	I	N	G	H	A	N	D	S	I	M	A	D	E	A	T	I	N	Y	B	R	E
4	A	C	H	I	N	T	H	E	U	P	P	E	R	L	E	F	T	H	A	N	D	C	O	R	N	E	R	A	N	D	T	H	E	N	W	I	D	E	N	I	N	G
5	T	H	E	H	O	L	E	A	L	I	T	T	L	E	I	I	N	S	E	R	T	E	D	T	H	E	C	A	N	D	L	E	A	N	D	P	E	E	R	E	D	I
6	N	T	H	E	H	O	T	A	I	R	E	S	C	A	P	I	N	G	F	R	O	M	T	H	E	C	H	A	M	B	E	R	C	A	U	S	E	D	T	H	E	F
7	L	A	M	E	T	O	F	L	I	C	K	E	R	B	U	T	P	R	E	S	E	N	T	L	Y	D	E	T	A	I	L	S	O	F	T	H	E	R	O	O	M	W
8	I	T	H	I	N	E	M	E	R	G	E	D	F	R	O	M	T	H	E	M	I	S	T	X	C	A	N	Y	O	U	S	E	E	A	N	Y	T	H	I	N	G	

The decrypted text can now be read across the rows.

It should be pointed out that originally the idea that the binary numbers were 5 digits long was arrived at via trial and error. It wasn't until Elonka reminded me that John Wilson had drawn our attention to the fact that that altered horizontal spacing defined a group of 5 letters. While we had no appreciation of the significance of that at the time, in hindsight, it becomes a likely indicator of the length of the binaries.

This relatively simple solution for Part 3 made previous solutions of Part 1 and 2 look *much* too difficult in comparison. This inspired me to revisit those solutions.

Recovering the Key Words for Parts 1 & 2

The table is clearly a broad hint as to the method used to encrypt Parts 1 & 2 (*with the ABCD... index across the top and side a possible dead end, as they are not used in the decryption process*). But, to put these sections at a level of difficulty (*level of simplicity?*) similar to that of Part 3, Kryptos must contain some mechanism for conveying the key words PALIMPSEST and ABSCISSA to the operative in the field.

A likely place to hide those key words was in the table itself, possible by making a slight alteration to one letter in each line. Unfortunately, due to resolution limitations and jpeg artifacts, the photos currently available are not sufficient to make a definitive determination. However, they do provide *some* evidence that the serifs on letters corresponding to the key words have been shortened. Compare the P on line A (*where the first letter of PALIMPSEST would be expected to be located*) with the P on row B. The portion of the serif extending to the left appears to be slightly shorter.



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Because most photos of Kryptos were taken from a distance, and anyone taking close up shots tended to focus on the “message” side of Kryptos, there are very few good photos of the table side to work with. A photo in “Atomic Time” (*the companion book to Jim Sanborn’s new installation that opened at the Corcoran Gallery of Art in late October 2003*) shows several letters that appear to be consistent with this theory.¹ The blow up above shows the last T in PALIMPSEST. (Note that the shortened serif on the right of the T is actually on the left, as this photo is a front view of Kryptos in which the message side reads correctly, but the table is seen from the rear.)

Because of the shadows, curvature of the panel, and limited resolution of the photos, it is not possible to state as an absolute fact that this theory is correct. The best that can be said at this time is that the photos are *consistent* with this theory. It may take an actual inspection of the sculpture to confirm this speculation.

¹ *Atomic Time*, Jonathan P. Binstock – ISBN 0-88765-072-5. This book retails for \$29.95 and is available at the Corcoran’s gift shop. Corcoran Gallery of Art, 500 17th Street NW, Washington, DC 20006

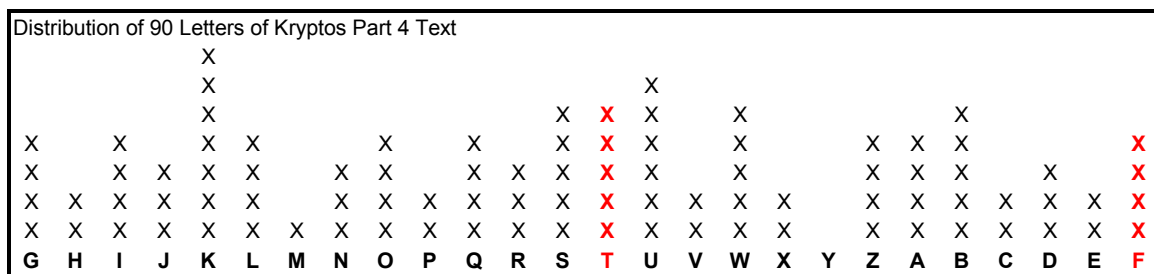
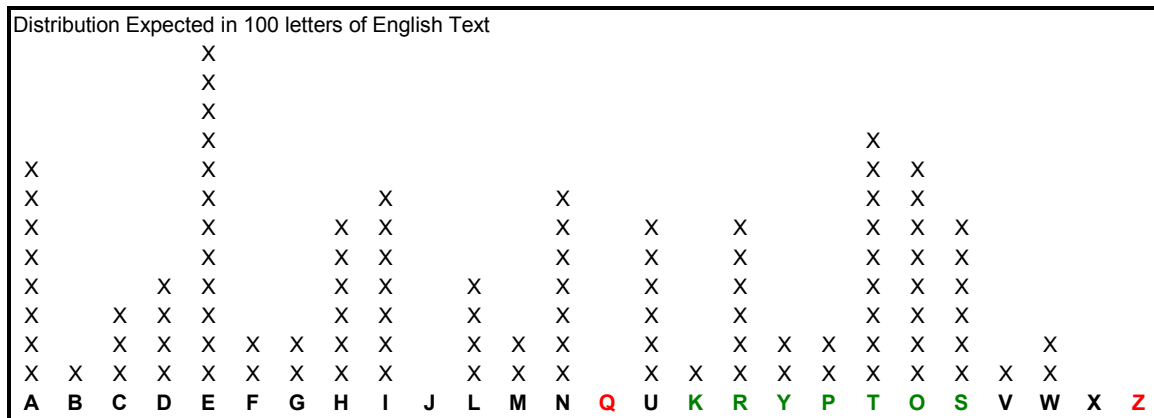
Speculation on Part 4

In light of the apparent simplicity of Parts 1, 2, and 3, it was time for a fresh look at Part 4. It has been widely speculated that Part 4 was a combination of a substitution and a transposition.

Some time ago I found a simple transposition for Part 4 that provided an interesting frequency match using the following alphabets:

ABCDEFGHIJKLMNOPQRSTUVWXYZ
GHIJKLMNOPQRSTUVWXYZABCDEF

This match up was derived by trial and error using strips of paper containing various alphabets, with and without the letters KRYPTOS, combined with each letter's corresponding frequency distribution in English and in the Kryptos text. The best match, after the strips were reassembled, is shown below.



The frequency match is absolutely stunning. And the fact that one of the alphabets contained the word KRYPTOS provided added evidence that this was the correct substitution. My speculation was that the few anomalies, the higher than expect number of Qs and Zs, could be a deliberate attempt to hide the frequency distribution from the casual observer – another dead end. Any measurement of the IC (*Index of Coincidence*) would indicate that a simple mono-alphabetic substitution was not used, preventing anyone from even attempting to look for a match of frequency distributions.

This Q & Z theory was given some credibility by a comment Scheidt made when asked if obscure languages were used in Kryptos. As usual, he did not give a direct answer. But the implication was clearly that foreign languages were considered, but ruled out. Almost as an afterthought he added that it was possible to use text that “didn’t look like English.”

The question now becomes – is there any set of possible hidden clues that support the selection of these alphabets for the substitution?

The GHI... alphabet immediately brought to mind Roger Anderson's recent observation that the GHI line on the table contained an extra character at the end.² Roger had already been doing some investigation regarding the significance of the alphabet on this line.

Maybe that extra L is an indication to the operative that this is one of the alphabets to be used in the substitution. However, my GHI... alphabet did not contain KRYPTOS. Are we dealing with another misleading dead end? Was the operative to use the L to find, not the alphabet, but as an indicator the alphabet should be shifted enough so that it starts with GHI...?

This only made sense if there was some other set of instructions that would allow the operative to construct the ABC... alphabet as well, including the position of the letters KRYPTOS.

Looking at that alphabet,

ABCDEFGHIJLMNQK**RYPTOS**V**WXZ**

I noticed the QU and immediately thought of the "errors" in Parts 1 & 2. I had always thought of these errors as the K & R errors, because those are the letters that show up in Kryptos. However, the errors in the plain text are Q and U. Could these errors be the markers that locate KRYPTOS within the ABC... alphabet?

While this does seem possible, unfortunately, that question can't be answered for certain until a complete solution is found for Part 4.

However, there is some additional (*and quite circumstantial*) evidence that the suggested substitution is indeed correct. But, before I get to that, you may have noticed that my frequency distribution charts used only 90 of the 97 characters from Part 4. That's because, in the search for other possible clues, I noticed the KR TOS YP in the cipher text.

?OBKR
UOXOGHULBSOLIFBBWFLRVQQPRNGKSSO****
TWTQS**JQSSEKZZWATJKLUDIAWINFBN**YP****
VTM**ZF**P**KWGDKZXTJCDIGKUHUAUEK**CAR****

It is possible (*I'd argue, likely*) that these letters provide a clue applicable to the transposition phase of Part 4. They could be a key word in a columnar transposition or some indication of the way text is to be written in to, or taken out of, a grid.

With these letters removed the letter count becomes 90. Add back the leading ? that was once thought to belong to Part 3 and the count becomes 91 – a very attractive number, as it limits any full rectangle to being either 13 x 7 or 7 x 13. (*An alternative would be to look at the KR SO YP as a solid block to be removed. Assume the T at the beginning of the 3rd line was a random occurrence and really does belong in the cipher text. The ? becomes either a divider between sections or the question mark at the end of Can you see anything Q? The count remains at 91.*)

² Note that this extra character is not shown in most transcriptions of the Kryptos sculpture (include those in the group's files or links sections). To correct any existing documents, add an L to the end of the line on the table side beginning with "N GHI..." That line will now end with "...HIJL"

One of the reasons a 13 x 7 grid is interesting concerns the cipher text after the completion of the substitution. It becomes either:

?ISUIYI ABUFSNI FCZSSRZ FMKLLJM HAENRQL NDLNNXE TTROQDE
FUWCORC HZSHKQQ GTZJERA WETYQDV WCAEUBU OUXEVOM

or

ISUIYIA BUFSNIF CZSSRZF MKLLJMH AENQRQL NDLNNXE TTROQDE
FUWCORC HZSHKQQ GTZJERA WETYQDV WCAEUBU OUXEVOM

In either case, there are two pairs of Qs that are separated by exactly 13 letters. Thus, if the text is written into a 13 x 7 grid, across the 13 columns and taken out by rows, those Qs end up paired.

If, going back to an earlier assumption, Qs are used as nulls to hide the English frequency distribution, we might hope to find 4 or 5 Qs grouped together at the end of the text. Getting them into pairs is at least a start.

Now consider that the KR SO YP might indicate that letters are written in to, our taken out of, our grid two at a time, and an interesting possibility develops:

A string of 5 Qs, taken 2 at a time would result in a lone Q and 2 pairs of Qs, which is exactly what shows up in the text.

Testing Part 4 Speculations

Is there any way to test the proposition that we have the correct substitution AND that a 13 x 7 grid is a good starting point for the transposition.

I propose that a suitable test would be to look at the unigram, bigram and trigram error statistics for the **cipher text**, **substituted text prior to a 13 x 7 transposition**, and **substituted text after a 13 x 7 transposition**.

The **cipher text** should bare no relationship to real English. Unigram error rates should be poor. Bigram and trigram stats should not change substantially when subjected to the 13 x 7 transposition, as it would be random text in and random text out.

The **substituted text** should show an improved Unigram stat, indicating text that more closely resembles English. Since it is still in the incorrect order, bigram and trigram error rates should show little or no improvement.

In the **substituted text after a 13 x 7 transposition**, the unigram stat would, be definition, remain unchanged. However, we would predict that some of the letters are now paired up with their partners. Thus bigram stats should improve. If the write-in (or out) pattern involves 2 letters at a time (as with KR SO YP), we would expect to see no improvement in the trigram stats, as no trigram would have remained intact. On the other hand, if KR SOT YP was used we should see some small improvement in the trigram because at least some of the letters were handled in groups of three.

It should be pointed out that these tests were designed prior to running the statistics to avoid any subconscious attempt to make the proposed test fit the data. Crank, Version 0.1.3 was used to carry out the transpositions and calculate that statistics listed in the following table:

? CIPHER TEXT	Unigram	Bigram	Trigram
Cipher Text	0.033201	0.016420	0.012432
Substituted	0.016055	0.015261	0.012372
Substituted and 13 x 7 Grid	0.016055	0.013109	0.012053
Substituted and 7 x 13 Grid	0.016055	0.016491	0.012370

As can be seen, the Unigram stats improved as predicted. In addition the bigram showed improvement in only the 13 X 7 grid – again, all as predicted. The trigram showed a very minor improvement.

An additional control test was done by placing the substituted text into a 7 x 13 grid. As would be expected there was no improvement in either bigram or trigram figures, clearly indicating that it that the cipher text should be written in across the 13 dimension rather than the 7.

The tests were then repeated with the T text (*question mark removed, T inserted*).

T CIPHER TEXT	Unigram	Bigram	Trigram
Cipher Text	0.032327	0.016036	0.012302
Substituted	0.017381	0.015148	0.012248
Substituted and 13 x 7 Grid	0.017381	0.013728	0.012054
Substituted and 7 x 13 Grid	0.017381	0.015650	0.012196

Which text, the ? or the T is correct? Although the results are too close to provide a definitive answer, several aspects point to the ? text being the correct version. The stats are *slightly* better. But more importantly, the very small improvement in the trigram error is consistent with use of KR SOT YP, which would have moved at least a small number of the letters as groups of three.

A Final Speculation

One last bit of off-the-wall speculation: A 13 x 7 grid implies that we might still be looking for an additional clue, a 13-letter key to be used in a columnar transposition. Elonka Dunin, after her dinner with Sanborn, remembered him saying that he put a great deal of effort into working on the right words for Part 1. Part 1 is 13 words long, making BSSATAOFLTNOI (*the first letter of each word*) an attractive candidate for a 13-letter key.

Summary

At this point we have:

1. Two candidates for the cipher text after substitution, with the ? version appearing somewhat more likely.
2. A high probability that a 13 x 7 grid is involved in the transposition.
3. Evidence that KR SOT YP provides some type of pattern for that transposition.
4. Speculation that the words in Part 1 supply a 13-letter key word for the transposition.

A Postscript

As I was writing this, a new member to the group, Patrick Foster, came up with an anagram using PALIMPSEST and ABSCISSA that could be Sanborn's final, somewhat sarcastic, message to us after it is all solved:

P.S. IT'S AS SIMPLE AS ABC.